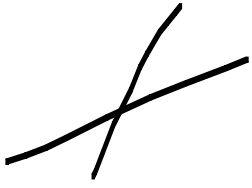


Unit 2.7 The derivative of a function.

Note Title

2014/03/02

pp 152-160



$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

f' is a new function of x .

derivative of f
geometrically the slope of the tangent line in the point x .

Example: Find $f'(x)$ if $f(x) = \frac{2}{x} = 2x^{-1}$

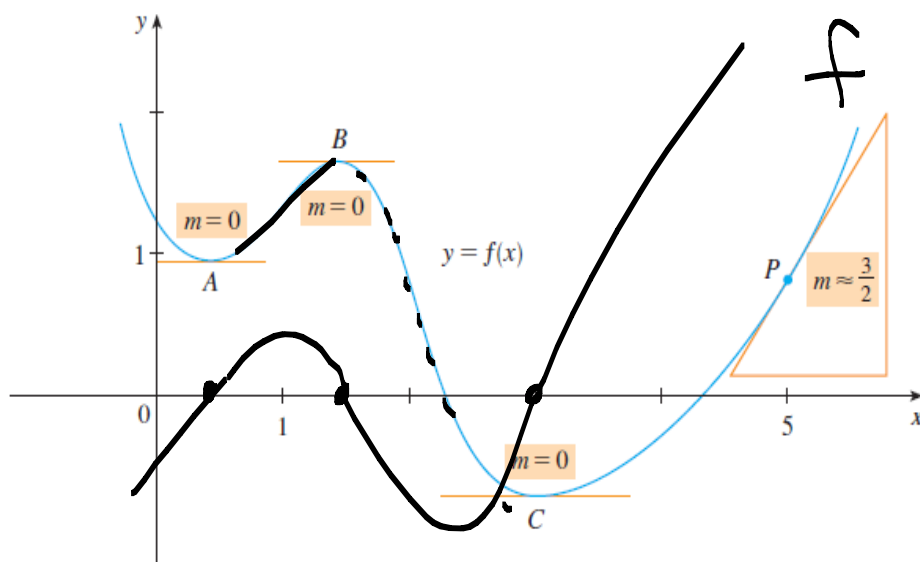
$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \left[\frac{\frac{2}{x+h} - \frac{2}{x}}{h} \right] \\ &= \text{SELF} = -\frac{2}{x^2} \end{aligned}$$

$\left(\frac{0}{0} \right)$

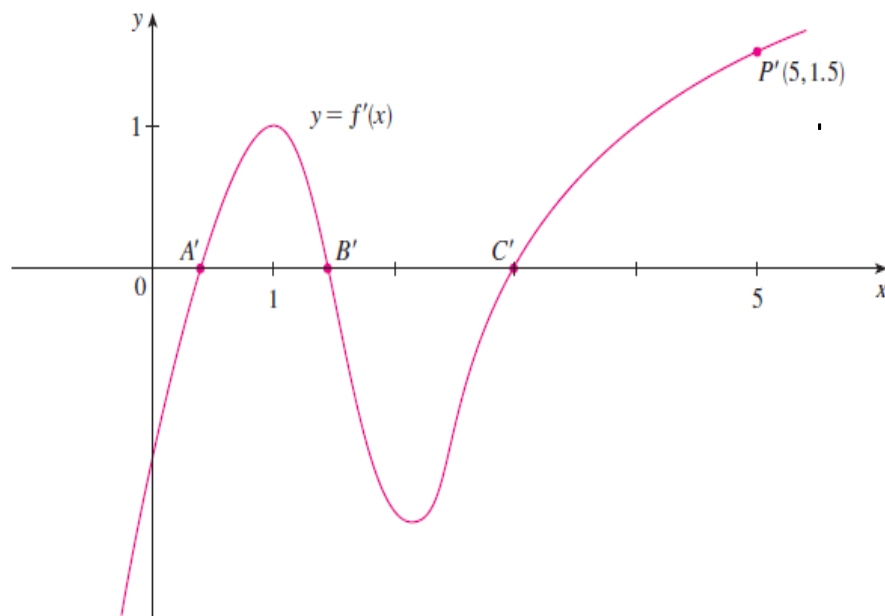
$\begin{cases} f(x+h) = \frac{2}{x+h} \\ f(x) = \frac{2}{x} \end{cases}$

(Check!!)

Geometric meaning



$f' ??$



Geometric meaning :

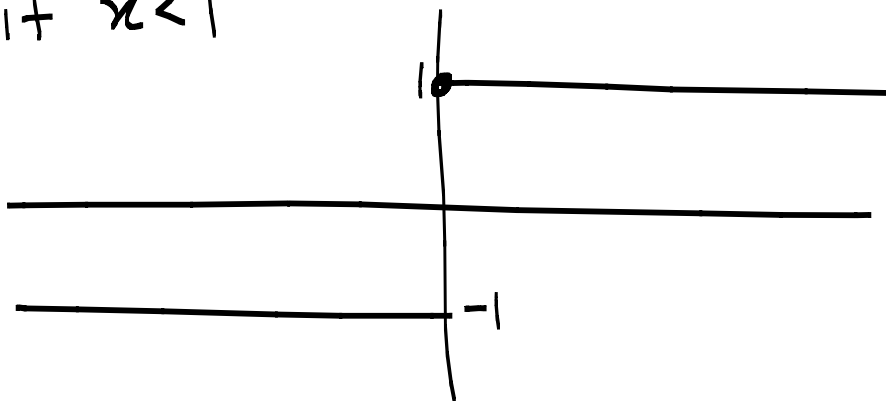
- $f'(x) = 0$ (horizontal tangent line)
- $f'(x) < 0$ (f is decreasing)
- $f'(x) > 0$ (f is increasing).

3 Definition A function f is differentiable at a if $f'(a)$ exists. It is differentiable on an open interval (a, b) [or (a, ∞) or $(-\infty, a)$ or $(-\infty, \infty)$] if it is differentiable at every number in the interval.

Is a function always differentiable?

Example: ① $f(x) = \frac{|x-1|}{x-1} = \begin{cases} 1 & \text{if } x \geq 1 \\ -1 & \text{if } x < 1 \end{cases}$

$$|x-1| = \begin{cases} x-1 & \text{if } x \geq 1 \\ -(x-1) & \text{if } x < 1 \end{cases}$$

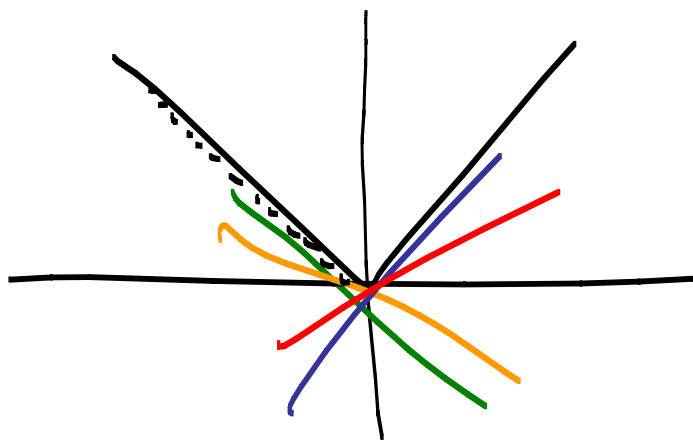


f is discontinuous at $x=0 \Rightarrow$
 f is not differentiable at $x=0$.

② $f(x) = |x|$
 f is continuous?

$$f(x) = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

$$f'(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases}$$

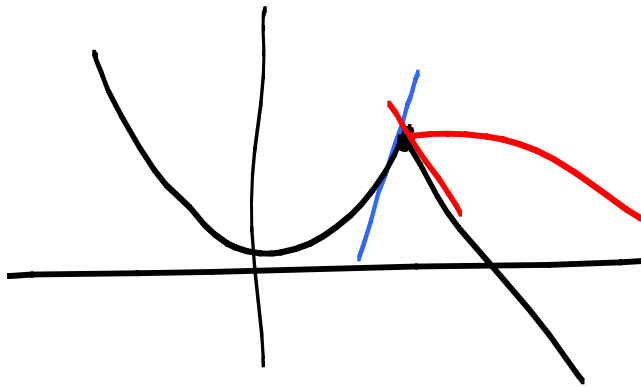


f is continuous on $\mathbb{R}$, but f not differentiable at $x=0$!

How can a function fail to be differentiable?

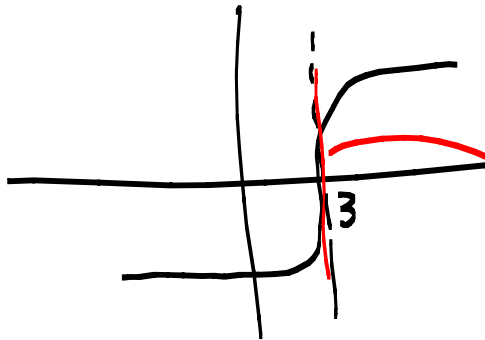
$$f(x) = \begin{cases} x^2 & \text{if } x < 2 \\ -x+6 & \text{if } x \geq 2 \end{cases}$$

①



corner (not smooth)

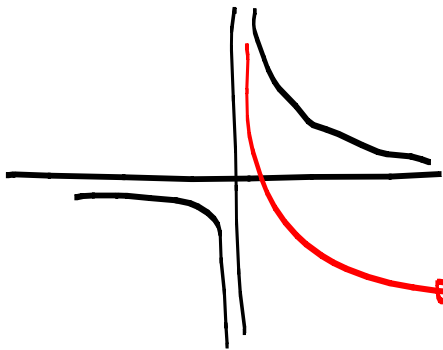
②



$$f(x) = (x-3)^{1/3}$$

vertical tangent at $x=3$.

③



$$f(x) = \frac{1}{x}$$

discontinuous at $x=0$

Notations

$$y = f(x)$$
$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx} f(x)$$
$$= Df(x) = D_x f(x)$$

D and $\frac{d}{dx}$ — differential operators.

Higher derivatives: p 160

$$\frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d^2 y}{dx^2} \quad \text{second derivative}$$

$$f''(x) \quad f'''(x) \quad D^3$$

$$\frac{d^n y}{dx^n} \quad f^{(n)}(x) \quad \frac{d^n}{dx^n} f(x)$$

Theorem 4 p158 NB

f differentiable at $x = a \Rightarrow$
 f continuous at $x = a$.

To prove: $\lim_{x \rightarrow a} f(x) = f(a)$

Given: $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$
exists!

Proof: LHS = $\lim_{x \rightarrow a} f(x) - f(a) + f(a)$

$$= \lim_{x \rightarrow a} \left[\frac{f(x) - f(a)}{x - a} \times \frac{x - a}{1} + f(a) \right]$$
$$= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \times \lim_{x \rightarrow a} (x - a) + \lim_{x \rightarrow a} f(a)$$
$$= f'(a) \times 0 + f(a)$$
$$= f(a) = \text{RHS}$$

f differentiable $\Rightarrow f$ continuous.

Is the opposite true i.e. does

f continuous $\Rightarrow f$ differentiable?

No!! Counter example: $f(x) = |x|$ is
continuous on $\mathbb{R}$,
but not differentiable at $x = 0$!